Lesson 2 · How machines learn from examples · for your pupils

Be the machine

40 minutes | Ages 10–16 · adapt for other ages below | No electricity needed | No devices needed | Photocopy-friendly

Big question

How can a machine learn from examples?

Pupils will be able to...

- Put the five steps of machine learning in order.
- Find a hidden pattern in labelled examples and use it to predict new ones.
- Explain how a spam filter learns from labelled messages.

You need

- Board and chalk (or large paper on the wall)
- Your plan of 18 shapes to draw (see Before class)
- Worksheet: Teach the machine (one per pair, optional)

Before class

Secret pattern: zogs have three corners. Plan 6 zogs, 6 not zogs (squares, circles, stars) and 6 test shapes, including an upside-down triangle. Vary size, shading and direction in both groups, so only the corners matter.

Time	What you do	What you can say	Pupils
Warm-up 0–5 min	Draw a ripe and an unripe mango on the board (or hold up real ones). Ask pupils how they would teach a younger child to tell them apart.	*"Would you give the child a rule, or show many mangoes and say 'ripe' or 'not ripe'?"*	Share ideas; notice that showing labelled examples works well.
Teach 5–13 min	Draw the five steps with simple icons: collect examples → label them → train: find patterns → test on new examples → use: make predictions. Use a spam filter as the example.	*"A spam filter learns from many messages labelled 'spam' or 'not spam'. Then it is tested on new messages before people use it."*	Copy the five steps; say them to a partner in their own words.
Zog game 13–27 min	Draw 6 zogs and 6 not zogs in two columns. Pupils study them silently for 2 minutes (training). Then draw the 6 test shapes one by one; pupils show 1 finger for zog, 2 for not zog. Keep score.	*"You are the machine. Study the examples, find the pattern, then predict each new shape. Keep your pattern secret for now."*	Study the examples, predict each new shape, and count how many they got right.
Pair talk 27–34 min	Pairs share the pattern they found. Take three ideas, then reveal the secret: zogs have three corners. Ask which shapes were hardest and what extra examples would help.	*"Which examples helped you most? What new examples would help a machine that got this wrong?"*	Compare patterns; suggest more, and more varied, examples.
Wrap-up 34–40 min	Link the game to the five steps: the labelled drawings were the training data, and the new shapes were the test. Then give the exit ticket.	*"You just followed the same steps a spam filter follows: labelled examples, a pattern, a test, then predictions."*	Complete the exit ticket.

Check understanding
Listen for “examples”, “label”, “pattern” and “test”. If a pupil says the machine “just remembers”, draw a brand-new shape and ask: “You never saw this one — how did you decide?”

Exit ticket
Write the five steps in order, and one thing a spam filter learns from.

Common mix-ups

Pupils may think…	Help them see…
The machine just remembers the examples.	Show a brand-new shape. Pupils who found the pattern can still sort it — that is what testing checks.
A prediction is always about the future.	Ask: “Is this message spam?” That is a best guess about now. Predictions can be about now or later.
AI keeps learning from everyone who uses it.	Many models stay the same until people retrain them with new examples. Some tools are updated more often.

Adapt it

Younger pupils
Ages 6–9: use real objects instead of drawings — leaves, bottle tops or seeds — with an easy pattern such as “zogs are round”. Play in small groups.

Older pupils and university
Pairs invent their own secret pattern and train another pair with eight examples. Discuss how many examples were needed. Use the words training data, label, model and testing.

Large classes
Pupils vote with one or two fingers so you can scan the room quickly. Row leaders collect their row’s score and best pattern idea.

No board or no space
Draw the shapes on the ground outside with a stick, or on scrap paper laid out in two groups.

Short on time? Do the warm-up, the five steps and the zog game with four test shapes (about 20 minutes).

What you need to know

People collect examples and label them, and the computer finds patterns in them to build a model. We test the model on new examples, then use it to make predictions.

Training data
The examples a computer system learns from — for example, thousands of emails marked “spam” or “not spam”, or leaf photos labelled by experts.

Label
The answer that goes with an example, usually added by a person — such as “spam” or “not spam”, or the name of a leaf disease.

Model
What training produces: the patterns the computer found, stored as numbers. It takes a new example and gives a prediction, such as “probably spam”.

Testing
Checking a trained model on new examples it has never seen. We already know their labels, so we can count how often its predictions are right.

My notes — how did it go?